HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Analyzed and reviewed extensive environmental and historical data to identify potential soil and groundwater contamination across multiple project sites.
- Assisted in the design and execution of ground investigations, including site inspections, geological logging, and collecting soil and groundwater samples for laboratory analysis.
- Performed geochemical and environmental data analysis to assess land and water quality, establishing potential connections between contamination sources and environmental receptors.
- Contributed to the assessment of material suitability for reuse, recovery, or disposal, feeding into sustainable materials reuse strategies for infrastructure projects.
- Prepared technical reports, environmental assessments, and regulatory submissions, clearly articulating findings and recommendations to project stakeholders.
- Collaborated effectively within multidisciplinary teams, integrating geoenvironmental insights into broader engineering solutions for sustainable infrastructure development.
- Containerized Python-based data processing scripts using **Docker** and deployed them to **AWS Lambda** for efficient, scalable analysis of environmental datasets.

SKILLS

Environmental Science – Geoenvironmental Assessment, Contamination, Site Investigations, Geological Logging, Soil Sampling, Groundwater Sampling, Land Quality, Water Quality, Remediation

Engineering & Analysis – Geoscience, Hydrogeology, Environmental Engineering, Data Analysis, Groundwater Modelling, Technical Reporting, Regulatory Submissions, Sustainable Development

Programming & Scripting – Python, SQL, Shell Scripting

Cloud & Containerization – AWS (EC2, S3, Lambda, CloudWatch), Docker

Tools & Methodologies – Jira, Confluence, Git, Jenkins, Agile, Scrum, Linux

PROJECTS

Geoenvironmental Site Assessment Framework

- Developed a Python framework to automate the initial review of historical environmental reports, identifying potential contamination hotspots and guiding site investigation planning.
- Integrated **geochemical data analysis** modules that parse laboratory results, calculating key indicators for land and water quality and flagging deviations from regulatory standards.
- Designed a standardized reporting template in Python, which consolidates findings from data analysis and site observations into comprehensive environmental assessment reports for stakeholder review.

Groundwater Contamination Modelling Support

- Assisted in parameterizing a **groundwater model** by extracting and analyzing historical soil and groundwater sampling data, providing input for contamination plume simulation.
- Wrote **Python scripts** to process and visualize large-scale geospatial environmental data, aiding in the identification of potential contaminant pathways and receptors.
- Validated model inputs against site-specific geological logs and historical survey data, ensuring the accuracy of the simulation environment.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)